

My work focuses on problems in machine learning theory and graph theory. The main theme of my research is to **understand how dynamics affect information in learning and networked systems**. I have a broad interest in ML and networks, and in particular I have spent the last two years of my postdoc appointment studying foundations of LLMs [15] and studying how communication network structure impacts data integrity [19, 17, 21, 20, 25, 24, 26]. I also have studied models for DNA data storage [18], graph based private data retrieval [22], and interactive graph learning [10]. I develop and utilize tools from combinatorics, information theory, and dynamics to tackle interesting problems at the intersection of computer science, mathematics, and engineering. The rest of this research statement will outline the problems and techniques I’ve developed over my career, with particular focus on the following questions,

1. What are the foundations of self-attention in Large Language Models? I view self-attention as an interacting particle system [15]. I am interested in dimensionality reduction techniques and conservation laws that allow for explainability of these very complicated systems.
2. How is data quality impacted by the topology of a communication network? I have answered this question for the ‘gossip’ communication model [17, 19] by leveraging tools from graph theory and statistical physics. I have early results for this question when the network topology can vary over time [25, 26].
3. How many imperfect observations are needed to recover a combinatorial object? I am interested in algorithms and lower bounds for two concrete models of this question: reconstructing DNA from subsequences [18], and reconstructing automata from random walks [16].

1 Foundations of Large Language Models

In my PhD I focused on classical notions of learning (PAC learning, query learning, online learning etc.). Modern deep-learning algorithms, including LLMs, are very difficult to study under these frameworks since gradient descent on the loss landscape is a black-box. In lieu of this limitation, a recent line of work has formalized the inference-time behavior of LLMs from an interacting particle perspective [11]. Given the ubiquity of LLM usage in STEM, this research direction is very timely and provides avenues for explainability of these systems. The key feature that separates LLMs from other neural networks is the *self-attention layer*, which allows the LLM to extract long-distance correlations in various domains. It was recently shown that self-attention behaves as a non-linear ordinary differential equation (ODE) which is driven by the high-dimensional learned parameter matrices [11, 3]. This is the starting point for my current work.

Along with a number of collaborators, I have shown that linear self-attention¹ is in fact a Kuramoto-like system [15]. The Kuramoto model is a very important system used to explore phase transitions, synchronization, and chaos in populations of coupled agents [30] (you may have seen demonstrations of pendulums synchronizing on a wobbly table; this is the physical motivation for Kuramoto). The connection between self-attention and the Kuramoto model allows us to leverage powerful dimensionality reduction techniques, **explaining the long-time system behavior for any parameter matrices**. Previous work could only describe the long-time behavior for a very restricted class of parameters by applying gradient flow techniques [11, 6, 3]. LLMs work on a sequence of N tokens embedded in $\mathbb{R}^d$, leading to a coupled ODE in Nd variables. For $d = 2$ our study shows linear self-attention is essentially governed by a single complex-valued ODE – such an immense reduction in variables allows us to rigorously describe the long-time behavior of the system for arbitrary parameter matrices, even as $N \rightarrow \infty$. We are finishing follow up work which generalizes our theory to d -dimensions, again for linear attention. The high-level idea for the reductions is that linear-attention evolves tokens via a *pure second-harmonic coupling*, which preserves certain group structures leading to inherently lower-dimensional dynamical behavior.

Our paper also shows the linear self-attention dynamics can display a wide range of dynamical behavior in the long-time limit. We have identified settings when tokens converge to a pair of points $\pm x_\infty \in \mathbb{S}^{d-1}$, cluster to a pair with $\frac{d}{dt} \pm x_\infty(t) = c$, partially cluster, and neither cluster nor converge. Long-time behavior is related to the expressability of LLMs, so showing such rich behavior already in $d = 2$ is a key theoretical advancement in explainability of these systems. The analysis of the cases above is aided by numerical software that I wrote for identifying the dynamics based on phase portraits of the

¹Here ‘linear’ refers to the self-attention kernel used, the underlying ODE/PDE is still nonlinear.

reduced dynamics. Additionally, I have identified that there are parameter regimes where the system evolves as a **Hamiltonian flow**, which is very surprising given the previous literature has focused heavily on gradient flow methodologies. This opens a new research direction - identify parameter regimes of self-attention which admit energy conservation instead of energy minimization. In addition, I have work in preparation which studies the interplay between the self-attention and feed-forward layers in LLMs, and I've shown the existence of sharp transitions where feed-forward layers dominate the dynamics.

2 Gossip Networks

One of the simplest communication protocols used in real systems is known as *gossip*, in which each user randomly sends and receives updates from nearby users. This protocol uses very limited computation resources at each user, since it only requires local information of the underlying graph structure. A fundamental question for this local communication model is: how new is the information held by a given user at a particular moment? There is necessarily a tradeoff between communication overhead and information quality in networked systems; centralized protocols have high communication overhead but good information quality, while local protocols have the opposite tradeoff. The *age of information* (AoI) [14] is one measure of information quality that has seen a surge of interest in the information theory community. Informally the AoI of a packet of data measures how out of date the packet is by measuring the elapsed time since the packet's creation.

The model I consider represents the network as a graph where vertices exchange packets continuously in time according to independent Poisson processes [31, 32]. As vertices share data, they only store the most up-to-date data they have received so far. This ensures that the whole network is essentially a sensor for the 'state of the world', where separate nodes have imperfect but correlated tracking of the world. My goal is to understand how the overall network structure impacts the quality of sensing as measured by the AoI metric. This compliments other work, which optimizes the communication model while keeping the network structure fixed [23].

Combinatorial Approaches

The first works in the area considered the long-time average age in the complete graph and cycle graph [31, 32]. The proof technique utilized tools from control theory to express the AoI as a certain recursion on subsets of finite graphs. Using theory I developed, **I've strengthened this foundational result** [21] in a simpler proof with no reference to control, allowing these recursions to hold on locally-finite infinite graphs. The recursive formula is the starting point for many works in age of gossip [28, 5]. The main technical challenge for finding the AoI is that the recursion tree is exponential in size of the graph. I became interested in AoI when I attended a talk by Srivastava and Ulukus [27] where they bounded the recursion using properties of the 2-D square grid. Their proof contained hints of a connection between AoI and the connectivity of the graph. As a PhD student **I identified the Cheeger constant as a combinatorial measure** for AoI in many classes of graphs [17]. This led me to obtain tight bounds on the AoI in new graph families. In the same paper, I proved AoI is a monotone graph property (solving an open problem and justifying earlier results [29, 13]) and leveraged this fact to **prove a thresholding phenomena in the Erdős-Reyni random graph**. I've also identified graphs where small changes in the gossip protocol lead to large changes in the AoI [24].

Probabilistic Approaches and Dynamic Networks

Towards the end of my PhD I became interested in universal bounds for AoI – at that point no such result existed. My work [19] obtains such a bound, which relies on the *local structure* of the graph, as opposed to the global structure the Cheeger constant captures. We proved a connection between AoI and the statistical physics model 'first passage percolation' (FPP), which allowed us to **develop a novel combinatorial parameter tightly capturing the AoI in arbitrary graphs**. We leveraged this new connection to **solve numerous open problems** in the area. Notably we solved the following questions from the survey [12]: the AoI scaling on square lattices, random geometric graphs, a cycle + random matching, and found a family of graphs with arbitrary polynomial AoI scaling. In a followup work I used the connection to FPP to recover the recursive identity in [31] with a very short proof and no control theoretic machinery [21]. In particular **my derivation holds in a stronger setting** of infinite, locally finite graphs and the time limit can be replaced with a suitably large stochastic time. This new probabilistic framework allowed me to study AoI in time-varying networks, where the previous recursive constructions cannot be applied [26, 25, 20].

3 Reconstruction Problems

I have a general interest in problems of the form ‘how many noisy observations of a combinatorial object are necessary/ sufficient to recover the original object?’, which are discussed below.

String and Tree Reconstruction

The string reconstruction problem was introduced over 20 years ago as a simplified model for DNA data storage [2]. An arbitrary binary string S is passed through a deletion channel where each bit is removed with probability p — we call a (random) subsequence from this process a *trace* of S . The string reconstruction problem asks ‘how many traces are needed to recover an arbitrary string S with success probability $1 - \delta$?’ This is a notoriously hard problem, with the best lower bound polynomial in (n, p, δ) [7], while a recent breakthrough shows quasi-polynomial sample complexity [4]. A similar problem was asked in the case of trees [8], and I have proven reductions from tree reconstruction to string reconstruction. My first contribution was to show that **tree reconstruction is no easier than string reconstruction** [18]. This is somewhat surprising, since trees have richer structure we may have hoped to use their structure to obtain better sample complexity bounds. The idea of my reduction was to encode strings as a special type of tree, and show that the tree deletion channel acts essentially as the string channel on these special trees. As a converse, I showed there is a labeling of the vertices of any tree which allow it to be recovered via a coupling to the string reconstruction problem. Therefore, the problem of recovering labelled trees from traces is equivalent to the string reconstruction problem up to polynomial factors. Finally, I proved **identities for the deletion channel** itself, including a connection between bit probabilities and generating functions induced by strings.

Learning Finite Automata from Random Walks

Learning deterministic finite automata (DFA) has a rich history tracing back to the early days of learning theory as a model for autonomous robot navigation [1]. Early hardness results for passive learning were complemented by polynomial sample complexity bounds for active and query learning models. I’m interested in a generalization of a learning model proposed by Freund et al. [9]. In the model I consider, an agent performs a random walk on a DFA with random transitions, and random 0-1 labels on each state. The goal of the learning procedure is to eventually make no mistakes when predicting the state labels of the DFA. In the case of an unbiased random walk, [9] proved there is an algorithm which stops making mistakes after $O(n^5)$ steps in the random walk. It turns out that their algorithm does not generalize to biased random walks. A significant portion of my PhD thesis [16] is dedicated to constructing a new algorithm which learns when the random walk is extremely biased to taking 1-transitions. The main challenge in this setting is that very rare events contribute a significant number of errors in any algorithm. I have **constructed an algorithm that learns the core of the DFA** with polynomially many mistakes via an agnostic learning procedure. There are uncommon, but not very rare events which give the learner useful information that can be leveraged to build a representation of the most visited sites in the DFA.

Future Work

My current research program is primarily focused on the mathematics of LLMs, in particular from the interacting-particle perspective. My work provides a new framework for studying self-attention via dimensionality reduction techniques, and has raised a number of new directions I am keen to pursue. Perhaps the most natural is to generalize our theory past $d = 2$; this is the focus of an ongoing followup paper, where we have reduced the system to $\mathcal{O}(d)$ coupled ODEs. Additionally our approach ignores the feed-forward layers present in real-world LLMs, which is a common simplification found throughout the literature. The feed-forward layers act as a universal external force driving tokens, and I have a current work deriving necessary and sufficient conditions for the strength of this field to drive tokens to a clustered state. Our empirical studies suggest these conditions hold in real-world LLMs.

Beyond ongoing work, I wish to find parameter settings which lead to energy conserving/ Hamiltonian dynamics. This is an important theoretical advancement, since previous work [11] has focused on the gradient-flow perspective which is complementary to Hamiltonian dynamics. Additionally, I want to conduct numerical experiments to study transitions to chaos in the self-attention dynamics — these transitions have been observed in Kuramoto and I have some very preliminary evidence this also occurs in self-attention. Finally I want to investigate dynamic metastability for anisotropic parameter matrices. This essentially asks how and when short-time behavior is different from long-time behavior.

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